

EIB Group Training & OD Platform: Architecture & Operations Manual

1. Overview

This document serves as the master reference guide for the EIB Group Learning Management System (LMS) and 90-Day Strategic Plan portal. It captures the underlying architecture, deployment strategies, strict security hierarchies, and operational procedures.

2. Technology Stack

- **Framework:** Next.js 15 (App Router)
- **Styling:** Tailwind CSS & shadcn/ui
- **Database:** PostgreSQL
- **ORM:** Drizzle ORM
- **Authentication:** Better Auth

3. Deployment & Infrastructure

Cloud (Vercel)

- The application is currently deployed on Vercel and connected to a Neon Serverless Postgres database.
- CI/CD is fully automated; pushing to the **main** branch on GitHub triggers an instant deployment.

On-Premise (Docker / VM)

- A completely self-contained **Dockerfile** and **docker-compose.yml** are provided at the root of the project.
- **Services:**
 - **lms-app** (The Next.js application running on port 3000)
 - **db** (A secure, containerized PostgreSQL 15 database)
- **Deployment Command:** `docker compose up -d --build`
- **CI/CD Note:** To automate deployments on-premise, the IT department must configure a webhook or GitHub Action to execute the Docker build command on the VM whenever the **main** branch is updated.

4. Role-Based Access Control (RBAC) & Hierarchy

The Roles

The system operates on four strict permission levels:

1. **Learner (learner):** Can enroll in courses, take quizzes, and track personal progress.
2. **Subsidiary Manager (lead):** Can track their own team. They only see data for users who registered under their exact subsidiary.
3. **Directorate Head (lead with wildcard access):** A middle-manager who oversees multiple subsidiaries within a specific directorate (e.g., DCI).

4. **LMS Super Admin (group_head)**: The absolute highest level of access. Can see all data across the entire EIB Group.

The Subsidiary Hierarchy

Subsidiaries are grouped in the UI to securely manage sensitive organizational structures:

- **Group Leadership**: EIB Group, Directorate of Clandestine & Intelligence
- **Directorate of Clandestine & Intelligence (DCI)**: DCI - SAC, DCI - PSAP, DCI - RAW, DCI - Intel
- **Commercial & Operational Subsidiaries**: EIB Stratoc, Luftreiber Automobile, POCTOVA, Briech Atlantic, Briech UAS, Bright FM, BEF

5. Security & Access Codes

To prevent unauthorized access to high-level data, the following security checks are hardcoded into the system:

- **LMS Super Admin**: Restricted **strictly** to `michael.marquis@eibgroup.com`. No one else can register for this role, even if they guess an access code.
- **EIB Group Head Code**: Any generic Group Head registering under the "EIB Group" subsidiary must provide the code: `EIB-GH-2026`.
- **DCI Directorate Head Code**: The Director of DCI must provide the code: `DCI-GH-2026` to register and gain wildcard oversight of SAC, PSAP, RAW, and Intel.

The 90-Day Plan Lockdown

The 90-day strategic presentation pages (`/`, `/dashboard`, `/roadmap`, `/strategy`, `/input`) are sealed inside a secure (`admin`) route group layout.

- **Only `michael.marquis@eibgroup.com` can view these pages.**
- Any other user (staff, subsidiary manager, or even the CTO) who attempts to visit the homepage will be instantly and silently redirected to the `/lms` learning portal.

6. Database Operations

Runtime Initialization

Because build-time migrations fail in some isolated environments, the database tables are **created dynamically at runtime**. Every time an authentication request is made (via `app/api/auth/[...all]/route.ts`), the system ensures all necessary tables exist using raw `pg` queries. This guarantees the app will never crash due to a missing table on a fresh on-premise deployment.

Seed Data

The 17 exact courses and their respective metadata are seeded into the database dynamically. If the database ever needs to be manually reset or re-seeded (e.g., on a fresh VM), simply navigate to: `https://<your-domain>/api/db/setup?reset=true`

7. How to Update the System in the Future

Adding a New Subsidiary

To add a new company or sub-unit, simply open `components/auth-form.tsx` and add the new name to the `SUBSIDIARY_GROUPS` object. It will instantly appear in the dropdown and be fully integrated into the RBAC logic without requiring any database changes.

Editing Course Content

Course catalog metadata, lesson text, and quizzes are hardcoded as a single source of truth in `lib/lms-content.ts`. To change a price, duration, or a quiz question, simply update that file and commit the code.